Baumarkt-Buddy – Release Roadmap Proposal (MVP + 2 Phases)

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Purpose and how to read

We propose building Baumarkt-Buddy in **3 releases**: Release 1 is the 2-week MVP we build and test for the pitch; Releases 2 and 3 are our proposal for after the course. Each release lists its requirements, the design components needed to build them, and the tests that close the release (V-model exit gate).

Source: `requirements/baumarkt-buddy-product-requirements.md` v1.0 (68 items: 24 FR, 10 SR, 11 NFR, 7 DR, 14 release-blocking tests). Full detail per item: `Baumarkt-Buddy_Release_Planning.xlsx`.

How the releases were cut (top-down):

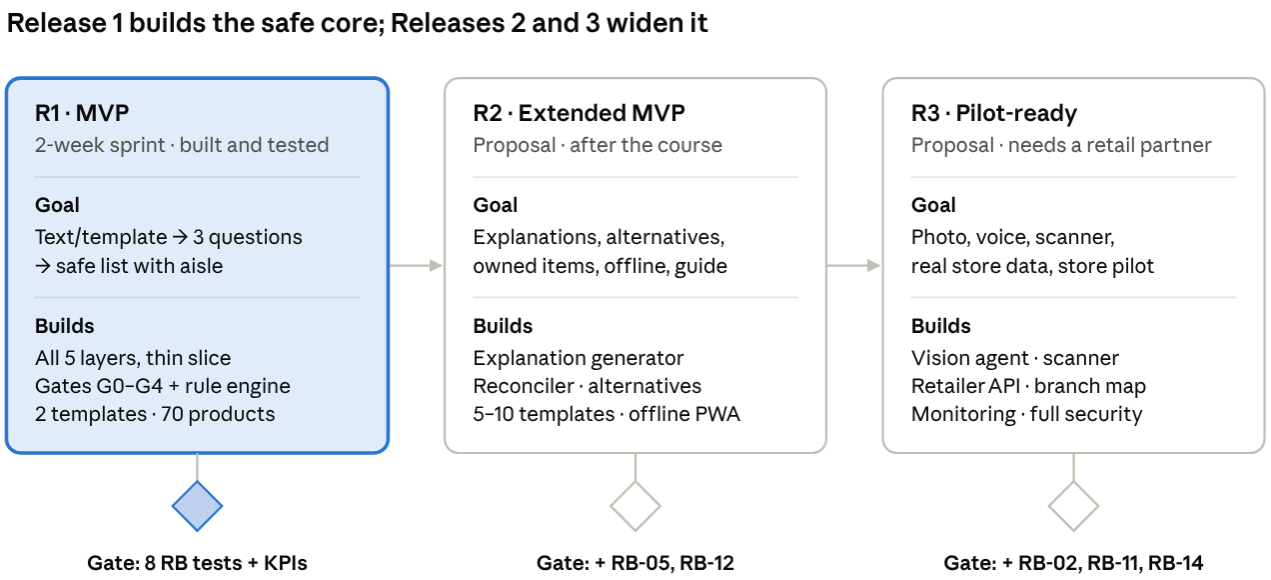
1. One goal per release decides what is in.
2. MoSCoW per release: **Must** = the goal fails without it · **Should** = important, not vital · **Could** = if time allows · **Won't** = later release.
3. Every safety requirement and release-blocking test that applies to a feature in the release is automatically **Must**.
4. Release 1 builds a thin slice through **all 5 architecture layers**; later releases widen layers, they don't add new ones.
5. A feature never comes before the data and components it depends on.

Architecture layers used in this document:

Layer	Role
L1 Presentation	Mobile web UI the customer sees
L2 Application	Orchestrator, project state, saving
L3 AI – proposes	LLM agents: classify, extract, ask, propose – never decide
L4 Control – decides	Deterministic code: gates G0–G4 and rule engine – decides safety, products, quantities
L5 Data	Catalogue, store map, templates, manufacturer data, evidence log

Layer	Role
Cross-cutting	Language, security & privacy, accessibility, monitoring, integration, review process

Roadmap at a glance



roadmap · 3 releases, 3 exit gates

Each release closes only when its release-blocking (RB) tests pass and its KPIs are met; earlier tests are always re-run. Release 1 has 43 requirements, Releases 2 and 3 have 10 each, and 3 stay out of scope.

Release 1 – MVP (2-week sprint, built and tested)

Goal: a customer types a DIY project or picks a template, answers up to 3 questions, and gets a safe, compatible shopping list with price, stock and aisle/shelf – only from our catalogue. **Use cases:** UC-1 hang a 20 kg shelf · UC-3 hang a picture/mirror · UC-5 high-risk request refused.

1.1 Features and requirements

Feature	Requirement IDs	MoSCoW	MVP scope
Describe the project in free text	FR-02 (text part), FR-03	Must	Text only, DE/EN; intent classified for 2 templates

Feature	Requirement IDs	MoSCoW	MVP scope
Confirm facts (load, wall type, size)	FR-04, SR-01	Must	Each fact shows status: observed / confirmed / unknown
Up to 3 follow-up questions	FR-05	Must	Skippable; missing wall type or load blocks fixing advice
Structured shopping list (BOM)	FR-06	Must	Materials, consumables, tools, PPE with quantity basis
Only catalogue products	FR-08	Must	Every line maps to a product ID in the CSV
Compatibility check	FR-09, SR-04	Must	Plug ↔ bit ↔ screw ↔ bracket; no adding of plug ratings
Price, stock, aisle/shelf	FR-11	Must	From mock CSV, labelled "mock"; no record = "ask staff"
Safety stops and hand-off	FR-15, SR-03, SR-06, SR-09	Must	High-risk refused; unknown facts stop the flow
Drilling and detector warnings	SR-02 (detector part), SR-05	Must	Warning on every drilling list
Manufacturer warnings preserved	SR-07	Must	Catalogue `safety_note` always shown
Recheck after changes	SR-08	Must	Changed load or wall type re-runs all checks
Usable on a phone in the aisle	NFR-01	Must	One-hand use, 375 px width
Fallback if AI fails	NFR-03	Must	Template list offered; no crash
Privacy and transparency	NFR-04, NFR-05	Must	No login; "You are using an AI assistant" notice
Ignore malicious input	NFR-07	Must	Text and product data never treated as instructions
Data separate from code	NFR-10, DR-01, DR-02, DR-03	Must	Existing `product_catalogue.csv` and `store_map.csv`
Project templates	DR-04	Must	2 templates: hang a shelf, hang a picture/mirror
Validated AI output	DR-05	Must	JSON schema check, max. 2 retries, then fallback

Feature	Requirement IDs	MoSCoW	MVP scope
Save and resume (session)	FR-01	Should	Session only; no account
Response time	NFR-02	Should	List in under 5 s
German default + English toggle	NFR-09	Should	UI texts in both languages
Basic security	NFR-06	Should	API key not in the browser; rate limit
Evidence record per check	DR-07	Should	Rule, inputs and time stored
Template review	SR-10	Should	Team review of mock templates and limits

1.2 Design and architecture components

Layer	Component	What Release 1 builds
L1	Project entry screen	Text field + 2 template buttons
L1	Fact confirmation view	Shows understood facts; user confirms or edits
L1	Clarification question UI	Max. 3 questions, skip allowed
L1	Shopping list view	BOM with price, stock, aisle, mock and AI labels
L1	Safety banner & hand-off	Refusal, blocked state, detector warning
L1	Mobile layout	One-hand layout, basic contrast
L2	Orchestrator	Fixed pipeline: G0 → intake → G1 → planner → rule engine → G2 → G4
L2	Project state machine	Draft, Needs clarification, Spec confirmed, BOM reviewed, Products validated, Safety blocked, Data unavailable
L2	Session persistence	Project kept for the session
L2	Revalidation trigger	Fact change invalidates the list
L3	Intake agent	Classifies intent, extracts candidate facts (text)
L3	Clarification agent	Phrases the questions in DE/EN

Layer	Component	What Release 1 builds
L3	Planner agent	Proposes BOM roles only, no products
L4	G0 High-risk filter	Electrical, gas, load-bearing, plumbing, roof → refuse
L4	G1 Required-facts check	Wall type and load confirmed before mounting output
L4	Rule engine R-01…R-16	Variants, quantities, compatibility, limits (mock values)
L4	G2 Catalogue check	Unknown product IDs blocked
L4	G4 Output schema validator	Retry max. 2, then "Please ask store staff"
L4	Input sanitiser	Prompt-injection guard
L4	Basket total	Sum and pack quantities; ambiguous pack → no total
L5	Product catalogue	Existing CSV (70 products)
L5	Store map	Existing CSV (aisle, shelf, quantity)
L5	Project templates	2 JSON templates with required facts and quantity rules
L5	Manufacturer data	Catalogue `safety_note` and load fields (mock)
L5	Evidence log	Rule, inputs, time per check
Cross	Labelling & i18n	AI notice, mock labels, DE/EN, metric, EUR
Cross	Security & privacy	API key server-side, no login, minimal data to the model
Cross	Review process	Team review of templates and rule limits

1.3 Verification and validation (exit gate)

- **Release-blocking tests (must pass):** RB-01 unknown wall blocks anchors · RB-03 wrong screw/variant flagged · RB-04 higher load revalidates · RB-06 no invented aisle · RB-07 detector warning · RB-08 manufacturer text wins · RB-09 injection ignored · RB-13 ambiguous pack → no misleading total.
- **Unit tests:** every rule R-01…R-16.
- **Validation KPIs:** 0 % invented products · 100 % high-risk refusals · list in under 5 s · at least 4 of 5 test users finish in under 2 min.

Capacity note: Release 1 has 43 items, 37 of them Must (mostly safety). If time runs short, cut to one template (hang a shelf) and drop the Shoulds – never the safety items.

Release 2 – Extended MVP (proposal)

Goal: make the plan richer and reusable – explanations, alternatives, owned items, offline checklist and assembly guidance across 5–10 templates. **Adds use cases:** UC-2 freestanding furniture · UC-4 paint a room. All Release 1 items stay and are re-tested.

2.1 Features and requirements

Feature	Requirement IDs	MoSCoW	Scope
Explain why each item fits and what is unverified	FR-10	Must	Built from the evidence log
Stale stock labelled with a verification path	RB-05 (with FR-11)	Must	Freshness rule per data source
Offline checklist readable without network	RB-12 (with FR-14)	Must	Stock shown as stale offline
Use items the customer already owns	FR-07	Should	Only confirmed suitable items reduce quantities
Cheaper / premium alternatives	FR-12	Should	Same rule engine; total updates
Offline project checklist	FR-14	Should	PWA, tick-off saved locally
Step-by-step assembly guide	FR-16	Should	Linked to BOM and reviewed manufacturer sources
Accessibility	NFR-08	Should	Screen-reader labels, scalable text, non-map list
Export and share a summary	FR-17, DR-06	Could	Link or PDF, selected info only
More templates	DR-04 (extended)	Must	5–10 templates; catalogue extended to ~150–300 products
Bilingual test suite	NFR-09 (extended)	Must	All calculations tested in DE and EN

2.2 New or extended design components

Layer	Component	What Release 2 adds
L1	Shopping list view	Explanations, alternatives, owned-item marks
L1	Checklist & offline cache (PWA)	Service worker, local save, stale labels
L1	Export / share module	Link or PDF of selected info
L1	Assembly guide view	Steps linked to BOM lines and sources
L1	Accessibility	Full screen-reader and contrast support
L2	Persistence	Local save beyond the session
L3	Explanation generator	Turns evidence records into plain-language reasons
L4	Owned-item reconciler	Deducts only confirmed suitable items
L4	Alternatives calculator	Finds compatible cheaper/premium options
L4	Rule engine	Rules for new templates; manufacturer system ratings
L5	Manufacturer / system data	Manuals, warnings, system ratings (mock until verified)
L5	Stock & price adapter	Timestamps and freshness / stale logic
L5	Templates and catalogue	5–10 templates, extended catalogue
L5	Evidence log	Source and version per record

2.3 Verification and validation (exit gate)

- Release 1 tests re-run as regression, plus **RB-05** and **RB-12**.
- **Validation KPI:** BOM completeness ≥ 90 % on the supported templates (expert check).

Release 3 – Pilot-ready (proposal)

Goal: prepare a controlled pilot in one store – new input modes (photo, voice, scanner), store search, verified store data, associate hand-off and operational monitoring. Requires a retailer partner and real data access.

3.1 Features and requirements

Feature	Requirement IDs	MoSCoW	Scope
Photo entry with role labels	FR-02 (photo part), FR-03 (extraction), SR-02 (photo part)	Must	Candidate objects only; never certifies wall, load or size
Photo without scale gives no exact size	RB-02	Must	Release-blocking
Location denied → postcode search works	RB-11	Must	Release-blocking
Changed product while shopping → revalidate	RB-14	Must	Release-blocking
Qualified expert review of templates	SR-10 (full)	Must	Before any public mounting guidance
Store search and retailer hand-off	FR-13	Should	Postcode or optional location; no implied purchase
Verified aisle/zone/rack per branch	FR-18	Should	Branch map with revision and time
Share summary with store associate	FR-22	Should	Selected info only
Monitoring and cost tracking	NFR-11	Should	Escalations, failures, latency, cost per project
Full security hardening	NFR-06 (full)	Should	Encryption, upload scanning, redacted logs
Pilot performance	NFR-02 (pilot)	Should	Multi-step plan under 60 s at p95
Scan products while shopping	FR-19	Could	Variant mismatch triggers recheck
Track assembly progress	FR-20	Could	Changed part invalidates later steps
Voice entry	FR-24 (voice part)	Could	Transcript shown and confirmed before use

3.2 New or extended design components

Layer	Component	What Release 3 adds
L1	Photo capture & role labelling	Upload with explicit role per photo
L1	Voice input	Speech-to-text, editable transcript
L1	Barcode / product scanner	Scan and compare with BOM variant
L1	Store search	Postcode and optional location
L2	Persistence	Backend storage for pilot projects
L2	Revalidation trigger	Triggered by scans and assembly changes
L3	Vision extraction agent	Candidate objects and labels from photos
L3	Intake agent	Accepts photo and voice input
L4	G1 Required-facts check	Photo facts stay "observed" until confirmed
L5	Stock & price adapter	Retailer API instead of mock CSV
L5	Store map / placement	Verified branch planogram
Cross	Retailer integration	Hand-off links, associate sharing
Cross	Observability & monitoring	Logs, dashboards, cost per project
Cross	Security & privacy	Full review; photo retention rules
Cross	Review process	Qualified expert approves templates and tests

3.3 Verification and validation (exit gate)

- All release-blocking tests except RB-10 pass (adds **RB-02**, **RB-11**, **RB-14**).
- **Validation KPIs:** pilot with 50–100 consented projects · exact variant precision ≥ 95 % · zero severe unsafe recommendations.

Architecture growth, out of scope and team decisions

How each layer grows

Layer	Release 1	Release 2	Release 3
L1 Presentation	Entry, fact confirmation, questions, list, safety banner	Explanations, alternatives, offline checklist, export, assembly guide	Photo, voice, scanner, store search
L2 Application	Orchestrator, state machine, session save, revalidation	Local save	Backend storage, scan-triggered revalidation
L3 AI – proposes	Intake, clarification, planner (text)	Explanation generator	Vision agent, voice/photo intake
L4 Control – decides	G0, G1, rule engine, G2, G4, sanitiser, basket total	Owned-item reconciler, alternatives, new rules	Photo facts kept "observed" until confirmed
L5 Data	70-product CSV, store map, 2 templates, evidence log	5–10 templates, ~150–300 products, manufacturer data, stock freshness	Retailer API, verified branch map
Cross-cutting	AI/mock labels, DE/EN, API key server-side, team review	Accessibility, bilingual tests	Monitoring, full security, retailer integration, expert review

Not in any of the 3 releases

Item	IDs	Reason
Approved reservation integration	FR-21, RB-10	Needs a retailer contract and API
Basket optimisation across retailers	FR-23	Multi-retailer data not available
Room visualisation	FR-24 (visualisation part)	Low value for the core problem

Item	IDs	Reason
Payment, checkout, native apps, indoor navigation	–	Excluded in requirements §3

Decisions for the team

- [ ] Approve the Release 1 scope (section 1.1), or cut to one template
- [ ] Confirm the 2 Release 1 templates: hang a shelf, hang a picture/mirror
- [ ] Confirm the mock safety limits for the prototype (1.25 × load, plasterboard ⩽ 25 kg, > 50 kg blocked)
- [ ] Choose the UI path: current static app in `app/` or Lovable
- [ ] Agree owners for each Release 1 component group (UI, AI, rules, data, tests)
- [ ] Accept Releases 2 and 3 as our proposal for the pitch